

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness for planning systems

Problem focus

Current agent training injects trajectories, feedback, and domain knowledge without asking whether the current step needs them.

FAST

direct action

SLOW

self-reflect

KNOW

retrieve knowledge

arxiv.org/abs/2504.03553

Audience: ML researchers & graduate students

Current Agent Training Uses “Flood Irrigation”

Indiscriminate knowledge injection can improve coverage but hurts efficiency and may introduce noise.

1

Trajectory imitation

Gold trajectories are injected as demonstrations.

Limitation: teaches behavior without situational control.

2

Trial-and-error

External feedback and corrections are added after

Limitation: costly loops and pattern collapse on unexpected inputs.

3

Knowledge-augmented

Domain knowledge is provided broadly or at every

Limitation: 100% knowledge usage raises cost and can distract planning.

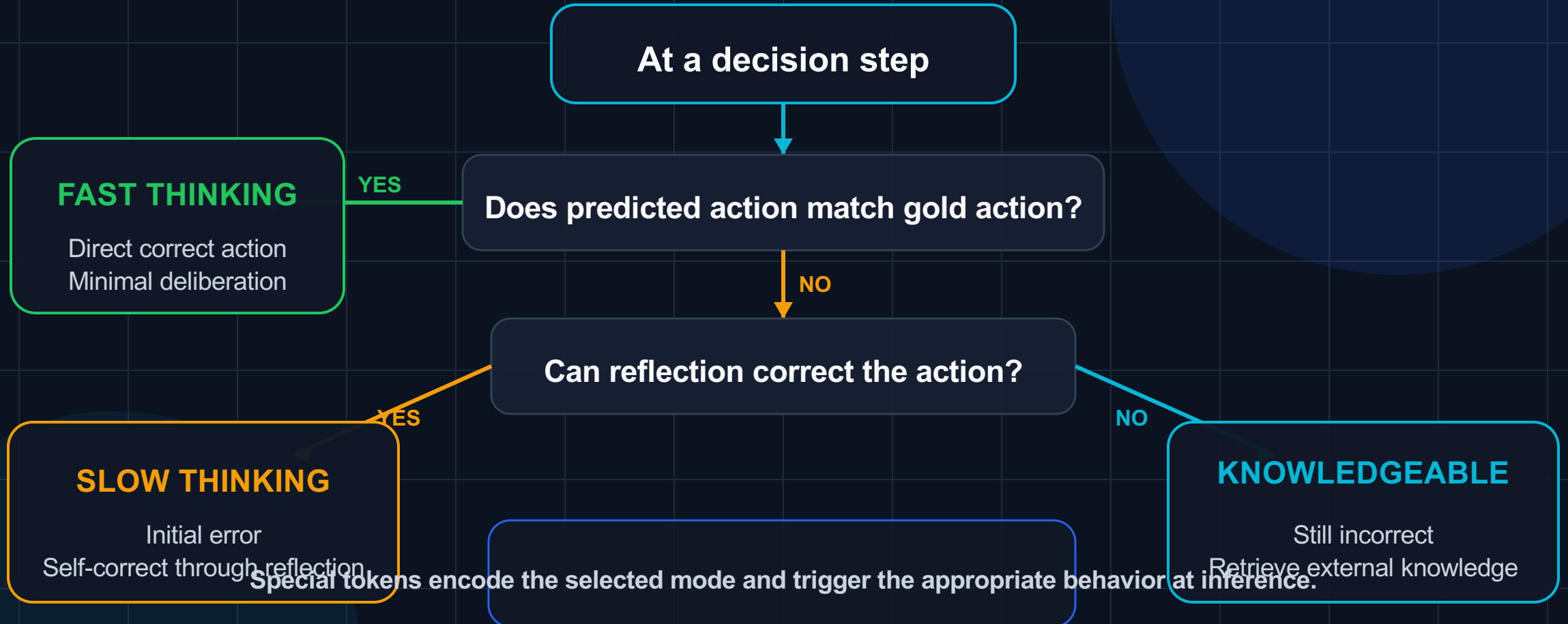


Human-like situational self-awareness

Can I handle this directly? Should I pause and reflect? Do I need external help?

Three Situational Modes

KnowSelf labels each decision step by whether the agent can solve it directly, after reflection, or only with knowledge.



KnowSelf Framework: Data → Training → Inference

A lightweight pipeline converts self-explored trajectories into self-aware planning behavior.



Figure 3 reference: data construction, learning, and deployment form a closed loop for regulating knowledge utilization.

Qualitative Examples: Reflection and Knowledge Rescue Planning

KnowSelf succeeds where strong reasoning models fail by choosing the right situational mode.

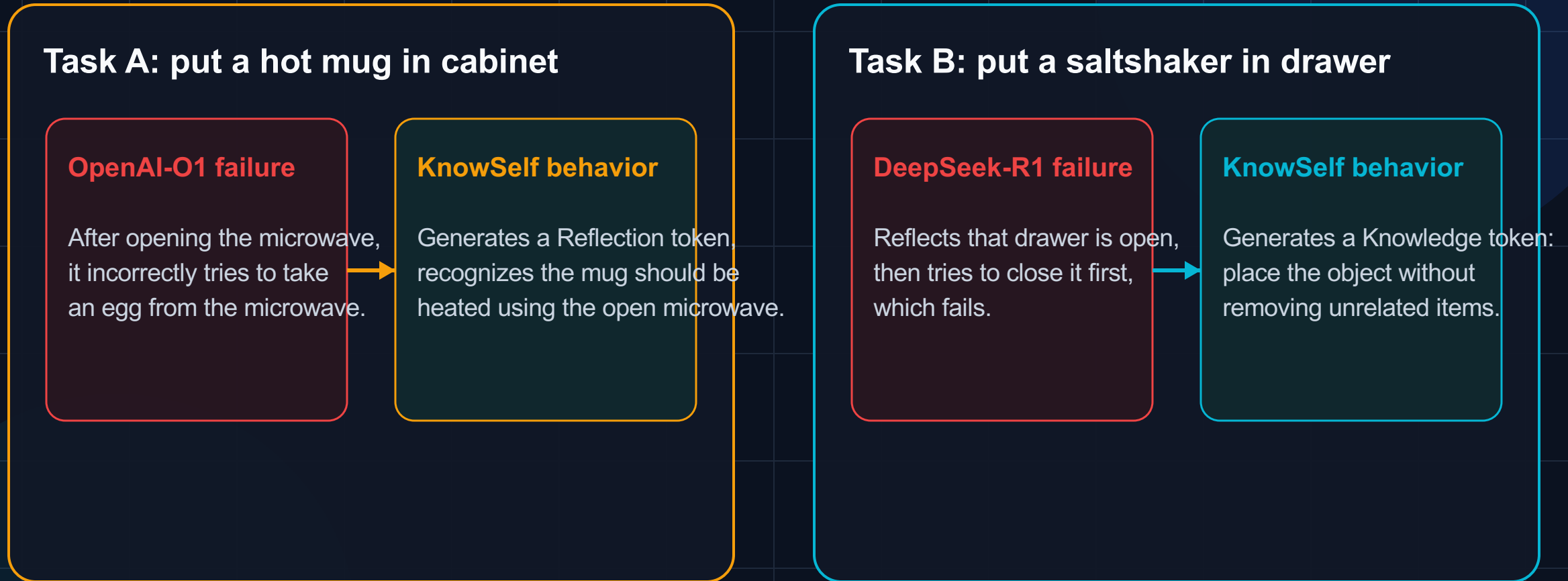


Figure 1 reference: qualitative comparison of KnowSelf vs. OpenAI-O1 and DeepSeek-R1.

ALFWorld Results: Accuracy by Backbone, Method, and Task Type

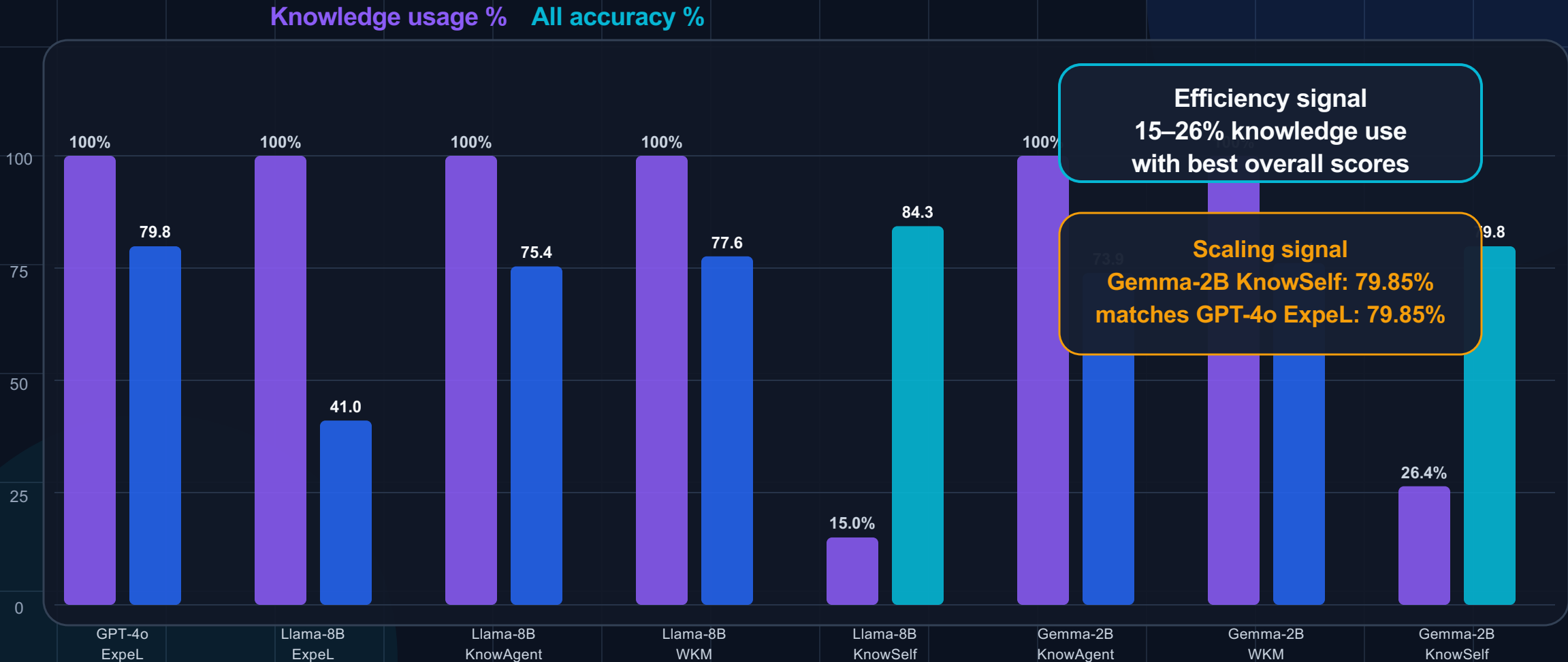
KnowSelf reaches 84.33% with Llama-8B using 15.01% knowledge; Gemma-2B reaches 79.85% using 26.41%.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Key contrast: selective knowledge beats blanket 100% knowledge use.

Minimal Knowledge, Maximum Impact

Efficiency view: KnowSelf uses a fraction of knowledge queries while matching or exceeding stronger full-knowledge methods.



Key Takeaways: Self-Awareness Is the Missing Piece

KnowSelf reframes planning agents around situational control of cognitive resources.

1

Blanket knowledge is wasteful

Indiscriminate injection can be suboptimal and costly.

2

Special tokens teach self-awareness

Heuristic labels identify fast, slow, and knowledgeable steps.

3

Strong empirical result

Llama-8B KnowSelf: 84.33 vs GPT-4o ReAct: 76.12 and GPT-4o ExpL: 79.85.

4

Efficiency advantage

Only 15.01–26.41% knowledge use versus 100% for competing knowledge methods.

5

Scales down effectively

Gemma-2B KnowSelf reaches 79.85, matching GPT-4o ExpL.

Future directions: richer self-awareness criteria, broader planning domains, and tighter integration with retrieval/memory systems.